MIL-W-8160D

17 MARCH 1961

SUPERSEDING
MIL-W-008160C(USAF)
7 APRIL 1959
MIL-W-8160B
19 JULY 1957

MILITARY SPECIFICATION

WIRING, GUIDED MISSILE, INSTALLATION OF, GENERAL SPECIFICATION FOR

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers the installation of wiring and wiring devices used for the interconnection of electric and electronic equipment in guided missiles.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

O-F-499 —Flux, Low Melting Point Silver Alloy Brazing.

QQ-S-561 —Solder, Silver.

QQ-S-571 —Solder, Lead Alloy, Tin Lead Alloy, and Tin Alloy; Flux Cored Ribbon and Wire, and Solid Form.

MILITARY

MIL-C-17 —Cables, Radio Frequency, Coaxial, Dual Coaxial, Twin Conductor, and Twin Lead.

MIL-C-71 —Connector, "N", for Radio Frequency Cables.

MIL-V-173 —Varnish, Moisture- and Fungus-Resistant, for the Treatment of Communications, Electronic and Associated Electrical Equipment.

MIL-C-572 —Cords, Yarns and Monofilaments Organic Synthetic Fiber.

MIL-I-631 —Insulation, Electrical, Synthetic-Resin Composition, Nonrigid.

MIL-T-713 —Twine and Tape, Lacing and Tying (for Use in Electrical and Electronic Equipment).

MIL-I-3190 —Insulation Sleeving, Electrical, Flexible, Treated.

MIL-C-3607 —Connectors, Pulse, for Radio Frequency Cables.

MIL-C-3608 —Connectors, BNC for Radio Frequency Cables.

MIL-C-3650 —Connectors, LC, for Radio Frequency Cables.

MIL-C-3655 —Connectors, Twin, for Radio Frequency Cables.

MIL-C-5015 —Connectors, Electric, AN Type.

MIL-W-5086 —Wire, Electrical, 600-Volt, Copper, Aircraft.

MIL-B-5087 —Bonding, Electrical (for Aircraft).

MIL-H-5166 —Handbook, Assembly, Service, and Maintenance Instructions (for Guided Missiles).

MIL-W-5274 —Wire, Electrical, Insulated, Aircraft.

FSC 1430

571063—61—(58)

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MIL-I-5585 —Installation of Low Pressure Oxygen Equipment in Aircraft, General Specification for.
MIL-E-6051 —Electrical-Electronics System Compatability and Interference Control Requirements for Aeronautical Weapon Systems and Associated Subsystems.
MIL-C-6136 Conduit, Flexible, Shielded, Aluminum Alloy.
MIL-S-6852 —Splice, Conductor, Electric, Disconnect.
MIL-S-6872 —Soldering Process, General Specification for.
MIL-W-7072 —Wire, Electrical, 600-Volt, Aluminum, Aircraft.
MIL-C-7078 —Cable, Power, Electrical, Copper, Shielded, Aircraft 600-Volt.
MIL-T-7099 —Terminals, Lug, Grip Style, for Aluminum Aircraft Cable.
MIL-W-7139 —Wire, Electrical, Insulated Copper, 600-Volt, 400° Fahrenheit.
MIL-I-7444 —Insulation Sleeving, Electrical, Flexible.
MIL-I-7798 —Insulation Tape, Electrical, Pressure-Sensitive Adhesive, Plastic.
MIL-B-7883 —Brazing of Steels, Copper, Copper Alloys and Nickel Alloys.
MIL-T-7928 —Terminals, Lug and Splice Crimp Style, Copper.
MIL-E-8189 —Electronic Equipment, Guided Missiles, General Specification for.
MIL-C-8384 —Connectors, Plug and Receptacle, Electrical (Molded Body) and Accessories, General Specification for.
MIL-S-8516 —Sealing Compound, Synthetic Rubber, Accelerated (for Electric Connectors and Electric Systems).
MIL-W-8777 —Wire, Electrical, Copper, 600-Volt, 150° Centigrade.
MIL-C-13777 —Cable, Special Purpose, Electrical.
MIL-W-16878—Wire, Electrical, Insulated, High Temperature.
MIL-C-25038 —Cable, Electric, Aircraft, High Temperature and Fire Resistant.
MIL-S-25063 —Specifications and Engineering Data for Airborne Electrical Equipment or Systems, Requirements for Preparation and Submittal of.
MIL-E-25366 —Electric and Electronic Equipment and Systems, Guided Missiles, Installation of, General Specification for.
MIL-C-26482 —Connectors, Electric, Circular, Miniature, Quick Disconnect.
MIL-D-70327 —Drawings, Engineering and Associated Lists.

STANDARDS

MILITARY

MS21919 —Clamp, Cushioned, Support, Loop-Type, Aircraft.
MS25123 —Terminal Board Assembly, Electrical.
MS25171 —Nipple, Electrical Terminal.
MS25281 —Clamp, Loop-Plastic, Wire Support.
MS35489 —Grommet, Rubber, Hot Oil and Coolant Resistant.

AIR FORCE-NAVY AERONAUTICAL

AN753 —Splice—Electrical Disconnect.
AN3054 —Nut, Coupling, Electrical Conduit.
AN3057 —Adapter—Electrical Accessory to Cable.
AN3420 —Bushing—Cable Adapter, Telescoping.
AND10106 —Tubing—Standard Sizes for Aluminum Alloy (52SO) Round.

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AND10380 --Fitting Installations—Standard AN Conduit.

PUBLICATION

ARMED SERVICES ELECTRO-STANDARDS AGENCY

ASESA 49-2B—Armed Services Index of R. F. Transmission Lines and Fittings.

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Conflicting requirements. In case of conflict between this specification and the detail specification for a particular guided missile or missile part, the detail specification shall in all cases prevail. Requirements of this specification shall not restrict the successful optimum performance capabilities of development missiles for which no adequate formal specification exists.

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with the order of precedence specified in MIL-D-70327 except as provided in 3.2.1 and 3.2.2.

3.2.1 *Commercial parts.* Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, such as screws, bolts, nuts, and cotter pins, having suitable properties may be used provided:

a. They can be replaced by the standard parts (MS or AN) without alteration.

b. The corresponding standard part numbers are referenced in the parts list and, if practical, on the contractor's drawings.

3.2.2 *Standard parts.* With the exceptions in 3.2.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers as specified in MIL-D-70327.

3.3 Materials and equipment. Materials and equipment shall be as specified herein and in the detail specification.

3.3.1 *Metals.* Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to missile propellants (solids, liquids, or fumes), salt spray, or atmospheric conditions as may be encountered in storage or normal service. Finish and coating shall be as specified in the detail specification.

3.3.1.1 *Dissimilar metals.* The use of dissimilar metals shall be as specified in MIL-E-8189.

3.3.2 *Nonmetals.* Nonmetals, including plastics, fabrics, and protective finishes, shall be moisture- and flame-resistant. They shall not support combustion nor give off noxious gases in harmful quantities and shall not be adversely affected by normal exposure to missile propellants (solids, liquids, or fumes), missile fluids, and weathering. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungistatic varnish conforming to MIL-V-173, or any other fungistatic agent approved by the procuring activity.

3.3.2.1 *Insulation tubing.* Insulation tubing for temperatures up to 100° C shall conform to MIL-I-631, type F, grade b, form U, class 1, category 1; MIL-I-7444, or shall be of a type approved by the procuring activity. Insulation tubing for temperatures greater than 100° C shall conform to MIL-I-3190 or of a type approved by the procuring activity.

3.3.3 *Equipment installation.* Electric and electronic equipment shall be installed in accordance with MIL-E-25366.

3.3.4 *Government-furnished equipment.* Wiring and wiring devices furnished by the Government shall be installed without modification, unless otherwise authorized or directed by the procuring activity.

3.3.5 *Modification.* The contractor shall not alter, rework, or modify wiring or wiring devices built to and meeting Government specifications, unless authorized or directed by the procuring

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activity. Such modifications shall be subject to Government inspection.

3.3.6 *Soldering.* Connections shall be soldered in accordance with MIL-S-6872 or the applicable detail specification. Solder used for electrical and electronic connections shall be in accordance with QQ-S-571 and of a suitable composition. High-temperature solder or other means shall be used where connections are made in high-temperature areas.

3.3.7 *Silver brazing.* Silver brazing shall be in accordance with MIL-B-7883.

3.3.7.1 *Brazing alloy.* The brazing alloy shall be in accordance with QQ-S-561, class 4.

3.3.7.1.1 *Brazing flux.* The brazing flux shall be in accordance with O-F-499.

3.4 Smoke and fire hazards. Wiring and wiring devices shall be so selected and installed as to minimise the danger of smoke and fire hazards in the missile. Protective means, both physical and electrical, shall be employed to provide reliability and safety commensurate with this requirement.

3.5 Ground return. For a grounded neutral or ground return system, the missile structure shall be considered the power system ground if practicable; if not, a ground cable may be used. Equipment that incorporates a ground terminal may utilize separate leads in the cable to permit grounding at one airframe contact if necessary; if not, the shortest lead shall be used. Equipment that does not incorporate a ground terminal and is internally grounded shall be grounded by the shortest lead if adequate contact to metallic ground cannot be made or maintained without corrosion at the mounts of the equipment.

3.5.1 *Bonding.* Bonding shall be accomplished as necessary to insure the performance of the missile and its components and to provide for safety during maintenance. Bonding shall be accomplished in accordance with the applicable requirements of MIL-B-5087.

3.6 Wires and cables.

3.6.1 *Selections.* Wires and cables shall be of types that are suitable for the application. Also, they shall be of such size as to insure adequate current capacity. The voltage drops shall be within the limits specified herein.

3.6.1.1 *Wire and cable for applications under 600V.* Unless otherwise specified (see 6.2), wires and cables shall be selected from the types conforming to MIL-W-5086, MIL-W-5274, MIL-W-7072, MIL-C-7078, or MIL-W-16878 for applications under 600V. For Army use MIL-C-13777 is applicable.

3.6.1.1.1 *Aluminum wires and cables.* Consideration shall be given to weight saving by use of aluminum wires and cables; however, approval for their installation shall be obtained from the procuring activity. Aluminum wires and cables shall not be attached directly to engine-mounted accessories installed in other areas of severe vibration nor installed where frequent connection and disconnection are required. All installation of aluminum wire and cable shall be relatively permanent. Aluminum wire shall not be used where the length of run is less than 3 feet, nor in areas where corrosive fumes exist.

3.6.1.2 *Wire for applications over 600V.* Unless otherwise specified (see 6.2), wires in accordance with MIL-W-5274, type III, class A, shall be installed for applications from 600 to 3,000V rms.

3.6.1.3 *Wire and cable for application in high-temperature areas.* Unless otherwise specified (see 6.2), wires and cables shall be selected from the types conforming to MIL-W-7139, MIL-W-8777, MIL-W-16878, or MIL-C-25038; however, if the design requirements can be met to better advantage with other types, wire and cable may be selected by the contractor in accordance with the order of precedence specified in MIL-D-70327.

3.6.1.4 *Coaxial cables.* Coaxial cables shall conform to MIL-C-17 or shall be listed in the Armed Services Index of R. F. Transmission Lines and Fittings (ASESA 49-2B).

3.6.1.5 *Minimum wire and cable size.* Wires and cables shall be of such cross section and temper as to provide ample and safe current-carrying capacity and mechanical strength. In gen-

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eral, wire and cable shall be not smaller than AN-22. When it can be demonstrated that the use of smaller wire and cable will result in no loss of performance and reliability and at the same time produce benefits, such as reduction in weight or improved accessibility, small wires or cables may be used if approved by the procuring activity. This provision is made specifically to permit the use of smaller wires or cables in bundles having a large number of wires or cables and having adequate support against adverse effects of vibration. For direct attachment to engine-mounted accessories, wire and cable sizes smaller than AN-18 shall not be used unless it can be demonstrated that no loss of performance and reliability will result.

3.6.1.6 *Current-carrying capacity.* The continuous-duty current in each wire and cable shall not exceed the corresponding value shown in table I. The use of this tabulation shall not eliminate other requirements for cable selection.

3.6.1.6.1 *Wire temperatures.* Table I current ratings apply to wires conforming to MIL-W-5086, MIL-W-5274, or MIL-W-7072 for a maximum wire temperature of 100° C with a maximum ambient temperature of 57.2° C. Insofar as practicable, table I current ratings shall also apply to approved high-temperature wires. When this is not practicable, derating methods compatibe with limiting insulation temperature and the ambient temperature of a given area may be utilized to achieve a savings in weight, provided substantiating data are submitted to the procuring activity for approval.

3.6.1.6.2 *Wires and cables in bundles.* Table I current ratings for wires and cables in bundles are based upon 15 or more cables carrying no more than 20 percent of the total carrying capacity of the bundle. In smaller bundles, the allowable percentage of total current may be increased as the bundle approaches the single-wire condition.

3.6.1.6.3 *Radio Interference.* Wires and cables shall be so selected and installed that the installation complies with the requirements of MIL-E-6051.

3.6.1.7 *Voltage drop.* The total impedance of wires and cables and ground return paths shall be such that the potential difference between the point of voltage regulation and the load does not exceed the limits shown in table II. Nominal system voltages shall be used for establishing the current to be used in determining the voltage drops.

TABLE I. *Current-carrying capacity of wires and cables*

Cable size		Continuous-duty Current--Amperes	
Aluminum	Copper	Single wire in free air	Wires and cables in conduit or bundles (see 3.6.1.6.2)
	AN-22	—	5
	AN-20	11	7.5
	AN-18	16	10
	AN-16	22	13
	AN-14	32	17
	AN-12	41	23
	AN-10	55	33
	AN-8	73	46
	AN-6	101	60
	AN-4	135	80
	AN-2	181	100
	AN-1	211	125
	AN-0	245	150
	AN-00	283	175
	AN-000	328	200
	AN-0000	380	225
AL-8		60	36
AL-6		83	50
AL-4		108	66
AL-2		152	82
AL 1		174	105
AL-0		202	123
AL-00		235	145
AL-000		266	162
AL-0000		303	190

TABLE II. *System voltages and allowable voltage drops*

Nominal system voltage	Maximum allowable voltage drop	
	Equipment operation	
	Continuous	Intermittent
28	1	2
115	4	8
200	7	14

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3.6.2 *Wire and cable identification (For Navy and Air Force).* To facilitate installation, assembly, and servicing in the field, each wire and cable installed shall be identified as specified herein. The identity of internal wiring of Government-furnished property shall not be altered by the contractor. The identification of wires and cables shall be accomplished by a combination of letters and numbers (wire identification code) marked on the installation wiring diagram and inprinted on the cable as shown on figures 1 through 4. Figure 4 is an example of cable identification coding as applied to typical wiring. The following requirements for identification shall apply:

a. Assignment of the cable identification code for wires or cables having functions listed under all circuit function letters, except Z, and for all wires or cables having functions listed under circuit function letter Z, except those used with Government-furnished property, shall be accomplished by the contractor (installing activity).

b. The cable identification code for wires or cables used with Government-furnished property having functions listed under circuit function letter Z shall be obtained from the procuring activity.

c. Wires installed to spare pins of connectors, such as potted connectors, shall be identified by the pin designation and wire size; e.g. D–16, a–12, or SMALL A–12.

d. Characters shall be of sufficient size to be legible and of permanent nature. The characters shall be of contrasting color; however, if black insulation is used, white characters are preferred.

e. The method of identification shall not impair the characteristics of the wire or cable.

f. Each wire or cable shall be identified at not more than 15-inch intervals throughout its entire length. Additions and exceptions to this requirement are as follows:

(1) Each wire except those described in 3.6.2.f(2) shall be marked as follows:

(a) The identification shall be marked on each wire within 3 inches of each termination and junction (except permanent splices) and at intervals of not more than 15 inches everywhere else. The identification shall be so located that shielding, ties, clamps, or supporting devices do not have to be removed in order to read the identification.

(b) The identification shall be printed on the wire.

(2) Wires as follows need not be marked in accordance with 3.6.2f(1), but shall be identified as specified.

(a) Each wire that is one of two or more wires and are included within a molded or woven (either metallic or nonmetallic) covering which was molded or woven in place on the wires. Each such cable shall have a sleeve on each end of it on which shall be printed the identification of each wire in the cable, the identification of the termination to which the wire is connected, and when there is any, the means of identifying the wire when the cable is opened (such as color coding).

(b) Each individually shielded wire that has no covering over the shield and is not included in a cable as specified in 3.6.2f(2)(a). Each such wire shall have a sleeve on each end of it on which shall be printed the identification of the wire at not more than 6-foot intervals.

(c) Each wire and cable that is less than 6 inches long. Such wires and cables need not be marked.

(3) Wires and cables less than 3 inches in length, such as jumpers, need not be identified in the missile but shall be completely identified on the drawing.

(4) For shielded, molded, or sealed cables, the identification code and color (if

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NOTES

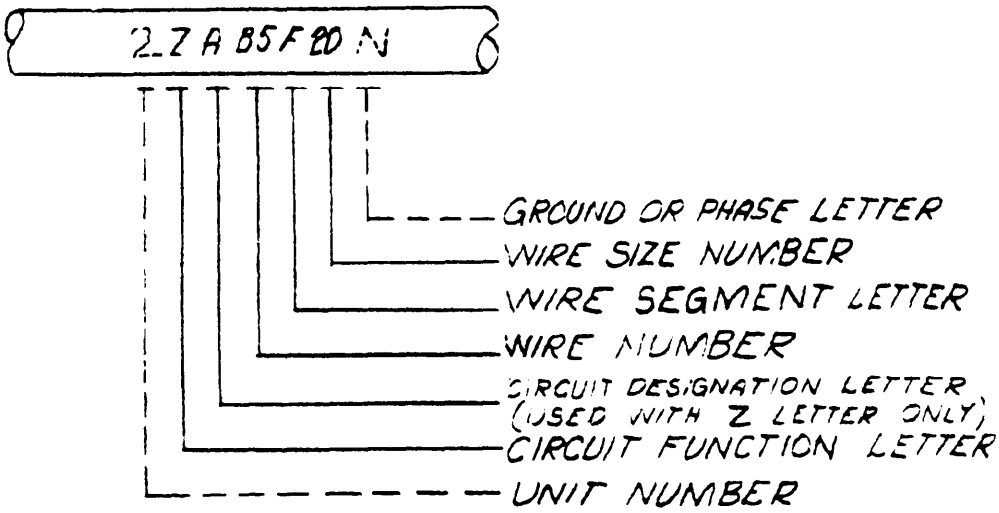
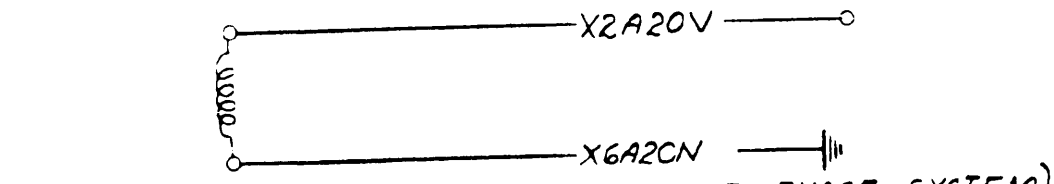


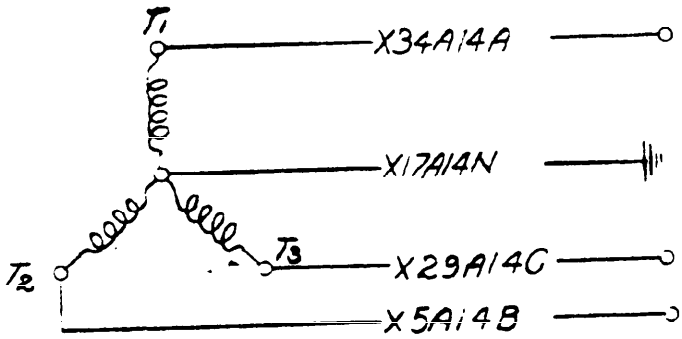
FIGURE 1. *Examples of wire identification coding.*

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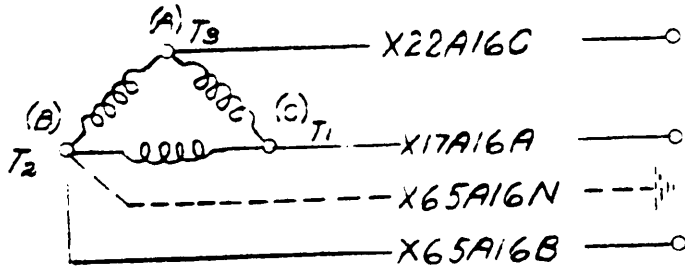
a (AS APPLIED TO POWER WIRING FOR SINGLE PHASE SYSTEM)

PHASE SEQUENSE
T_1-T_2-T_3



b. (AS REQUIRED TO POWER WIRING FOR THREE PHASE "Y" SYSTEM)

T_1-T_2-T_3
(C) (B) (A)



c. (AS APPLIED TO POWER WIRING FOR THREE PHASE DELTA SYSTEM)

FIGURE 2. *Examples of wire identification coding.*

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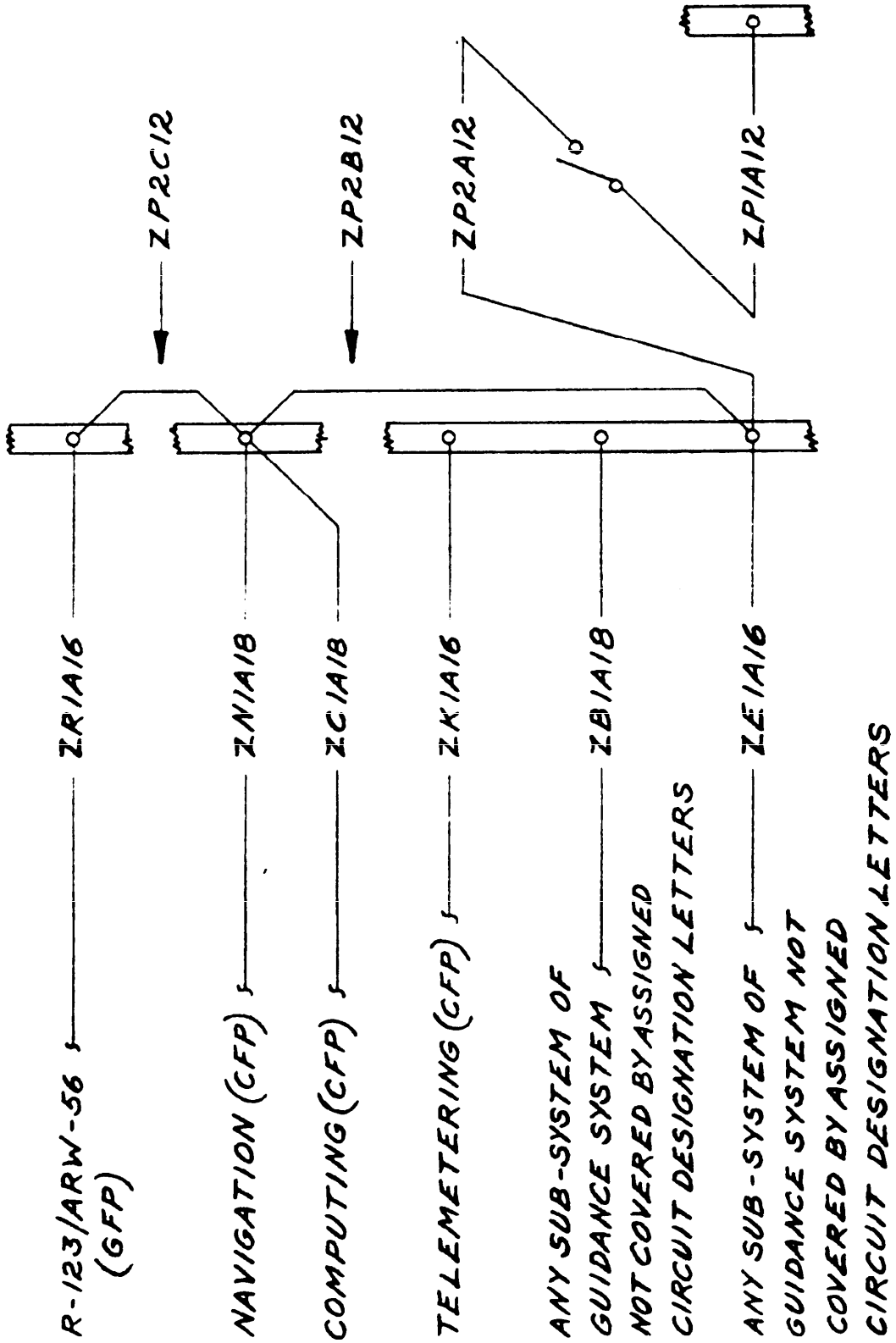


FIGURE 3. *Examples of wire identification coding.*

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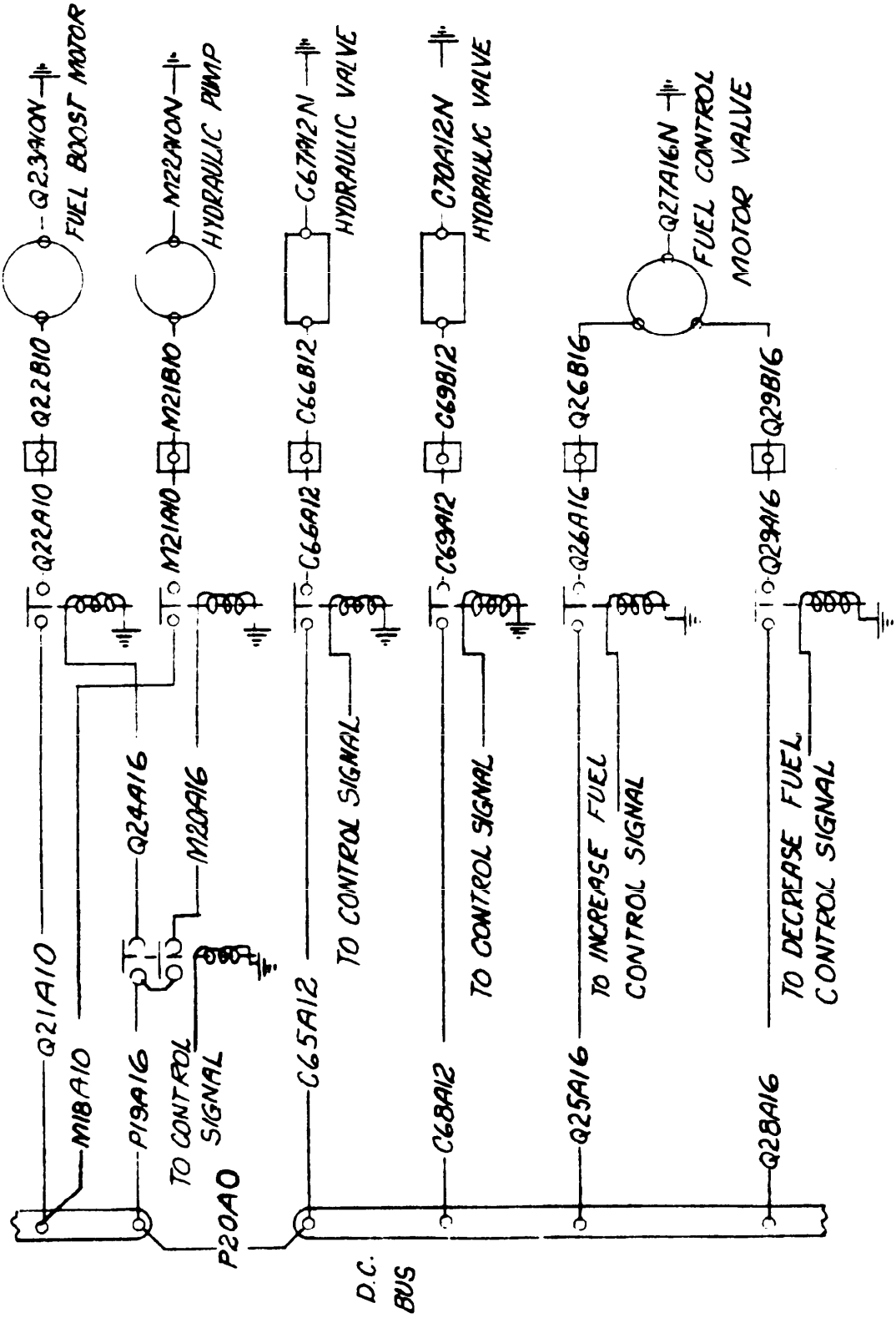


FIGURE 4. *Examples of wire identification coding.*

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applicable) of each conductor shall be placed on a sleeve outside the cable. Where not attached to connectors by molding or potting (making nonrepairable-type assemblies), each conductor shall be individually marked inside the connector.

3.6.2.1 *Code.* Reading from left to right, the wire or cable identification code illustrated on figure 1 is as follows:

a. Unit number—Where two or more identical items of equipment are installed in the same missile, the unit numbers 1, 2, 3, 4, et cetera, shall be prefixed to differentiate between wires or cables, or cables when it is desired that the equipment have the same cable identification. In order to facilitate interchangeability requirements, identical wiring located in left and right lifting surfaces, nacelles, and major interchangeable structural assemblies shall have identical cable identification and the unit number shall not be required.

b. Circuit function letter—The circuit function letter shall be part of the wire and cable identification as in table III. Typical circuits are listed under their respective circuit functions. Where table III is inadequate, additional information or assignments shall be requested from the procuring activity. Where a wire or cable is used for more than one circuit function, the circuit function letter of the circuit that is functionally predominant shall apply. When functional predominance is questionable, the circuit function letter for the wire or cable having the lowest wire number shall be used. It is recommended that categories of circuit function letters be established to cover flight test telemetering, radio, and radar gear which does not relate directly to guidance.

c. Circuit designation letter—The circuit designation letter provides a refinement of the circuit function specified in table III and shall be used only with circuit function letter Z. All other circuit functions shall not have a circuit designation letter.

d. Wire or cable number—The wire or cable number, consisting of one or more digits, shall be used to differentiate between wires or cables in a particular circuit. A different number shall be used for wires or cables not having a common terminal or connection. Numbers 900 to 999 inclusive shall be reserved for use by the procuring activity to identify wires and cables installed by service modifications.

e. Wire or cable segment letter—A wire or cable segment shall be a wire or cable between two terminals or connections. The wire or cable segment letter shall be used to differentiate between wire or cable segments in a particular circuit. A different letter shall be used for wire and cable segments having a common terminal or connection. Wherever practicable, wire or cable segments shall be lettered in alphabetical sequence and the letter A shall identify the first segment of each circuit starting at the power source. The letters I and O shall not be used as segment letters. Double letters AA, BB, et cetera, shall be used when more than 24 segments are required. Segments of wires joined by a two-way permanent splice shall have the same identical segment letter.

f. Wire or cable size number—The wire or cable size number shall be used to identify the AN or AL size of the wire or cable. For coaxial cables, the cable size number shall not be included.

g. Ground or phase letter.

(1) Unless otherwise specified by the procuring activity, ground cable letter N shall be used as a suffix to the cable identification code to identify any wire or cable that completes the circuit to the ground network. Such wires or cables shall be capable of being connected to the ground network of the aircraft

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TABLE III. *Function and designation letters*

Circuit function letter	Circuits
A	Armament
B	Photographic
C	Control surface
D	Instrument (other than flight or engine instruments)
E	Engine instrument
F	Flight instrument
G	Handling gear (when required)
H	Heating, ventilating, and de-icing
I	I shall not be used
J	Ignition
K	Engine control
L	Lighting
M	Miscellaneous
N	Unassigned (see note 1)
O	O shall not be used
P	D-C power or power-control system
Q	Fuel and oil
R, S, T, U, Y	Unassigned (see note 1)
V	D-C power and d-c control cables for a-c systems
W	Warning and emergency (in addition to those listed under other circuit functions)
X	A-C power The letters A, B, C applied as a suffix to the cable identification code shall indicate the phase sequence corresponding to the grounded phase. (Examples of wire identification coding as applied to a-c power wiring are shown on figure 2.)
Z	Guidance (see note 2) ZA Auxiliary assembly ZC Computing ZD Recording ZK Telemetering ZN Navigation (electronic and other) ZQ Special (light, Magnetic, et cetera, and combinations of circuit designations) ZR Receiving ZS Detecting and range, and bearing ZT Transmitting ZX Identification and recognition ZP Power (see note 3)

electrical system without causing malfunctioning of the circuit.

(2) Phase letters A, B, or C shall be used as a suffix on the table identification code to identify the phase of wires that are in the three-phase wiring of alternating-current systems. The phase sequence shall be A—B—C.

(3) Phase letter V shall be used as a suffix on the cable identification code to identify the ungrounded cable that is in a single-phase system.

3.6.2.2 *Coaxial cable identification.* In addition to the cable identification code, coaxial cables shall be identified within 3 inches of the end which terminates at the equipment to conform to the marking at the receptacle on the equipment. This shall not apply to coaxial cables that terminate in one connector with other wires or cables.

3.7 Junctions. Junctions should be kept to a minimum consistent with final field assembly requirements. Only approved devices, such as solderless-type terminals, terminal blocks, connectors, permanent splices, and feed-through bushings shall be used for cable junctions. Reliability factors, maintenance factors, and manufacturing procedures shall be considered in determining the need for and the type of device.

3.7.1 *Protection.* Where required, protection may be provided by covering the junctions, by installing them in junction boxes, and by locating them in such a manner that additional protection is not required.

3.7.1.1 *Hoods.* Terminal nipples conforming to MS25171 or suitable insulating hoods may be used to provide the required protection on terminal lugs and studs.

Note 1. Unassigned circuit function letters shall not be used until formally assigned by joint action of the services. Subject to approval of the procuring activity, unassigned circuit designation letters may be used to designate any subsystem of the guidance system not covered by assigned circuit designation letters.

Note 2. The cable identification code for wires or cables used with Government-furnished property which have functions listed under circuit function letter Z shall be obtained from the procuring activity. (Blocks of numbers for cable identification of circuit function letter Z will be assigned to each complete equipment by the Aeronautical Standards Group at the request of the procuring activity. The cable numbers assigned to each complete GFAE equipment will be utilized in the composite wiring diagram. Where necessary to use additional cables for interconnection, the cable numbers required will be obtained from the individual complete equipment block of numbers.)

Note 3. Where multiple guidance components are switched simultaneously or connected to a common junction point, the letters ZP shall be used for primary power leads from the common switch or junction point as shown on figure 4.

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3.7.1.2 *Nonmetallic covers.* Nonmetallic junction covers, such as fabric or plastic, shall maintain a high dielectric resistance and shall not absorb missile fluids. They shall be so installed as to allow for drainage, not interfere with mechanical movements, and in such manner that metal fasteners cannot cause short circuits. Nonmetallic covers in proximity with high-temperature equipment, such as resistors, shall withstand the maximum temperature without damage.

3.7.2 *Installation.* Electrical junctions shall be installed so that they are adequate both mechanically and electrically. They shall not be subject to mechanical strain nor used to support insulating materials. Junctions shall not depend upon insulators under compression for maintaining the connection tightness.

3.7.2.1 *Spacing.* Electrical junctions shall be adequately spaced to prevent detrimental leakage currents between circuits under the worst environmental conditions to which a missile will be subjected. Suitable insulating barrier material may be used to provide necessary creepage distance.

3.7.2.2 *Sensitive circuits.* Sensitive circuits shall be kept separate from other circuits at junctions when the proximity of such circuits has a detrimental effect on the performance of the equipment, system, or missile. This may be accomplished by using separate connectors for the sensitive circuits and by having at least one grounded terminal stud between sensitive circuits and other circuits on a common terminal block.

3.7.2.3 *Accessibility.* Junctions, other than permanent splices, shall be accessible for inspection and maintenance with a minimum of disturbance to other items and with the use of a minimum number and variety of special tools.

3.7.3 *Conductivity.* Electrical junctions shall be made in such a manner that the conductivity at junctions will be suitable for the circuit.

3.7.3.1 *Disconnect junctions.* Disconnect-type junctions, such as connectors and disconnect splices, shall have a conductivity of not less than that specified in MIL-C-5015, MIL S-6852 or MIL-C-8384, as applicable.

3.7.3.2. *Bolted junctions.* Bolted and permanent-type junctions, such as terminal blocks, shall be made in such a manner that the conductivity at the junctions will be not less than that of an equivalent length of cable equal in size to that making the junction.

3.7.3.3 *Aluminum junctions.* Special attention shall be given to aluminum wire and cable installations to guard against conditions that would result in excessive voltage drop and high resistance at junctions that may ultimately lead to failure of the junction. Examples of such conditions are improper torquing of nuts and inadequate terminal contact areas.

3.7.4 *Terminals and splices.* Wire and cable terminals and splices shall be of the crimp style, solderless type conforming to MIL-T-7928 for copper and MIL-T-7099 for aluminum, except where otherwise specifically approved by the procuring activity. For applications involving temperatures above the upper limit for materials used for preinsulated terminals and splices, the bare copper (type I) shall be used. Where class 2 terminals are used the installation shall be such that class 1 terminals will be adequate as a replacement without rework of the installation or terminal. Parts lists shall indicate the class 1 (MS or AN) terminal that may be used for service replacement of class 2 terminals, if installed. Where operating temperatures are above 450° F, other terminals may be used, subject to the approval of the procuring activity.

3.7.4.1 *Insulation.* Insulation tubing specified in 3.3.2.1 shall be placed over terminals where necessary to provide electrical and mechanical protection. Provisions shall be made for keeping the tubing in place. Wire identification may be printed on this tubing.

3.7.5 *Terminal blocks.* For ambient temperatures up to 250° F, terminal blocks shall conform to MS25123 for junctions of wires and cables requiring infrequent disconnection or the joining of more than two wires and cables to a common point. The number of wires and cables connected to any one stud shall be limited to four. The terminals shall be so installed that tightening the nut on the stud will not deform

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the stud. Equipment terminal studs may be used for joining wire and cable terminal lugs wherever they are suitable.

3.7.6 *Connectors.* Connectors shall be used for joining wires or cables or to connect to equipment when frequent disconnection is required to remove or service equipment, components, or cabling. The number of connectors shall be kept to a minimum.

3.7.6.1 *Adjacent locations.* Where similar connectors are used in adjacent locations, wires and cables shall be so routed and supported that improper connections cannot be made. If this requirement cannot be met, adequate marking or identification shall be provided in the form of color coding or tags. Maximum use shall be made of polarizing features or alternate insert positions to prevent misconnections.

3.7.6.2 *Drainage.* Connectors shall be so positioned as to permit fluids and condensate to drain out of and not into the connector. When connector locations do not allow positioning so that adequate drainage can be obtained, moisture-resistant connectors shall be installed or insulation tubing provided to prevent fluids from running into the connector.

3.7.6.3 *Connectors.* Connectors for missile wiring shall be in accordance with MIL-C-5015, MIL-C-8384, and MIL-C-26482 (for Navy use), or as approved by the procuring activity.

3.7.6.3.1 *Insulation.* When not otherwise adequately protected, insulation tubing shall be placed over each wire terminal connection in both plugs and receptacles. The tubing shall extend at least over the end of the terminal a sufficient length to enable tying or otherwise adequately securing. (This requirement does not apply to moisture-resistant type connectors.) Wire identification may be printed on this tubing.

3.7.6.3.2 *Adapters.* When not otherwise adequately supported, open wiring attached to AN connectors shall be supported by adapters conforming to MS3057 and bushings conforming to AN3420 or by an assembly consisting of a maximum length of 6 inches of insulation tubing, a set of approved ferrules, and a ferrule nut in accordance with AN3054, or by other approved means. Tightening the adapters or AN connectors shall cause no strain in cables or their connections. Adapters for special and miniature connections shall anchor the wiring securely under all conditions of stress or vibration.

3.7.6.3.3 *Special-purpose connectors.* Where installation conditions require pressurized, sealed, vibration-resistant, flame-barrier, fireproof or waterproof connectors, suitable special-purpose connectors in accordance with MIL-C-5015 or MIL-C-8384 shall be used.

3.7.6.3.4 *Moisture-resistant connectors.* Class R connectors in accordance with MIL-C-5015 and connectors which have been potted with sealant conforming to MIL-S-8516 in accordance with an approved method shall be used in applications where moisture resistance is necessary. The potting of connectors shall be subject to approval by the procuring activity.

3.7.6.3.5 *Bonding ring.* Where no ground pin is provided in an AN connector for terminating the shield of cable conforming to MIL-C-7078, cable shields shall be grounded directly to the structure, or as otherwise approved.

3.7.6.4 *Coaxial connectors.* Connectors for coaxial cables shall be in accordance with MIL-C-71, MIL-C-3607, MIL-C-3608, MIL-C-3650, MIL-C-3655, or shall be listed in the Armed Services Index of R. F. Transmission Lines and Fittings (ASESA 49-2B).

3.7.6.5 *Provision plugs.* When practicable, plugs that are in the guided missile for test purposes shall be secured by clamps or to dummy receptacles provided for the purpose, so that the plugs cannot swing on wiring and cause damage to themselves, wiring, or adjacent equipment. When dummy receptacles or clamps are not used, the method of stowage shall be such as to insure permanency of the installation, no damage to wiring or equipment, and no impairment of flight.

3.7.7 *Splices.* Splices shall be held to a minimum. Uninsulated splices shall be covered with suitable insulation tubing as specified in 3.3.2.1. The tubing shall extend at least ½ inch over each end, provide a snug fit, and be held

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securely in place. Preinsulated splices shall conform to MIL–T–7928. Splices shall be staggered to prevent shorts due to puncturing the insulation and to prevent excessive enlargement of the bundle.

3.7.7.1 *Permanent splices.* Permanent splices of a type approved by the procuring activity may be used for final assembly of subassemblies.

3.7.7.2 *Disconnect splices.* Disconnect splices conforming to AN753, or other approved disconnect splices may be used to facilitate removal of individual wires in nonvital circuits only upon specific approval of the procuring activity.

3.8 Junction boxes. Nonmetallic or metal junction boxes lined with insulation material are preferred to unlined, metal boxes in order to lessen the possibility of shorting between connections and the box.

3.8.1 *Construction.* Junction boxes shall be of suitable construction to house junctions, wiring ices, and equipment. The inside shall be of ght color to facilitate inspection and repair. e cover shall be fastened by approved devices and shall be captive by hinge or cord. If captive covers prove troublesome on small missiles, a deviation should be requested.

3.8.1.1 *Drainage.* Suitable holes, about 3/8-inch in diameter but not less than 1/8 inch, shall be provided at the lowest points of the box, except vapor-tight boxes, to allow for drainage with the missile on the ground, in launching attitude, and in normal flight attitude.

3.8.1.2 *Insulation.* The inside of metallic junction boxes shall be coated with an insulation material conforming to MIL–V–173 to minimize the possibility of shorts to ground.

3.8.1.3. *Identification.* Junction boxes shall be externally identified to facilitate correlation of the box with the wiring diagrams.

3.8.1.3.1 *Vapor-tight boxes.* Vapor-tight junction boxes shall be externally labeled VAPOR-TIGHT.

3.8.2 *Installation.* Junction-box wires and cables shall be adequately supported at convenient intervals to:

a. Provide neat and orderly arrangement

b. Provide ease of inspection and maintenance

c. Provide relief of strain on terminals

d. Minimize possibility of faults

e. Prevent vibration from damaging wires and cables or terminals.

3.9 Routing. Wiring shall be routed to:

a. Provide reliability

b. Not be source of interference or coupling between systems

c. Prevent malfunctioning of equipment by induction

d. Provide accessibility for inspection and maintenance

e. Minimize the possibility of damage.

3.9.1 *Antenna cables.* Each antenna cable shall be separated from any other antenna cable or antenna cable group.

3.9.2 *Engine-compartment wiring.* The wiring in an engine compartment, from the point of disconnection for removal of the engine, shall be interchangeable between all engine installations having identical equipment and for the series of missile.

3.9.3 *Protection.* Wires and cables shall be installed and routed so as to be protected against chafing and damage from fumes and fluids.

3.9.3.1 *Flammable lines and equipment.* Wires and cables shall be supported and separated from lines containing flammable liquids, gases, and oxygen, and from their associated equipment. For Navy and Air Force procurement, the distance between their supports shall be no greater than the separation from these lines and equipment except that the separation from oxygen lines and equipment shall be as required by MIL–I–5585. Wires and cables shall not be attached to lines and equipment containing flammable liquids and gases, unless they require electrical connections or clearance between them is less than 2 inches. When the clearance is less than 2 inches, the separation shall be maintained by attaching a cable clamp to a fitting on the equipment or a clamp on the line, and then no less than 1/2-inch separation shall be achieved.

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3.9.3.2 *Trapped fluids.* Wires and cables installed in locations where fluids may be trapped and the cables contaminated shall be properly routed and protected against fluids damage.

3.9.3.3 *Drip loop.* Where wiring is dressed downward to a connector, terminal block, panel, or junction box and when fluids or condensate may run into these devices, a trap, drip loop, or other equally effective means (such as sealing) shall be provided. This requirement does not apply to wiring kept in a controlled environment.

3.9.3.4 *High-temperature equipment.* Wires and cables shall be kept separate from high-temperature equipment, such as resistors, exhaust stacks, heating ducts, and de-icers, to prevent cable insulation deterioration.

3.9.3.5 *Additional Protection.* Electrically unprotected power and distribution circuit wires, cables, and buses shall be given particular mechanical protection, such as in the form of extra insulation. Bus centers shall be located within insulated enclosures so isolated as to prevent a line-to-ground or phase-to-phase short circuit that would disrupt the electrical power system.

3.9.4 *Slack.* Excess slack shall not be provided in wires or cables, except for drip loop and service requirements. However, enough slack shall be provided to:

a. Permit ease of maintenance including connector replacement
b. Prevent mechanical strain on the wires or cables, junctions, and supports
c. Permit free movement of shock- and vibration-mounted equipment.

3.9.5 *Radius of bend.* The minimum radius of bend for power distribution wires or cables shall be 10 times the outside diameter, except that at terminal strips where the cable is suitably supported, the radius may be 3 times the diameter of the cable. Where it is impracticable to install these wires and cables within the radius requirements, the bend shall be enclosed in insulation tubing.

3.9.5.1 *Coaxial cables.* For coaxial cables, the minimum radius of bend shall not unduly affect the characteristics of the cable.

3.9.6 *Wires and cables near moving parts.* Wires and cables attached to assemblies where relative movement occurs at hinges and rotating pieces, such as flight control surfaces, shall be installed or protected in such a manner as to prevent deterioration of the wires and cables caused by relative movement of the assembly parts. This deterioration includes abrasion of one wire or cable upon the other, and excess twisting and bending. As a design objective, bundles shall be installed to twist instead of bend across hinges.

3.10 Support. Wires and cables shall be supported by cushion clamps conforming to MS21919 or MS25281, grommets meeting MS35489, or other approved devices at intervals of not more than 24 inches, except when contained in ducts or conduits.

3.10.1 *Between support devices.* Devices used for supporting wire and cable shall be of an approved type which is replaceable by cushion clamps conforming to MS21919 or MS25281, grommets conforming to MS35489, tape in accordance with MIL-I-7798, and cord and string in accordance with MIL-C-572, type P, form C, or MIL-T-713, type P (waxed), class 2. Because of unusual environmental conditions, other approved types may be selected by the contractor as required.

3.10.2 *Size and type.* Supporting devices shall be of a suitable size and type so that the wires and cables will be held securely in place without damage to the insulation.

3.10.3 *Wire and cable installation.* Wires and cables shall be properly supported and bound to:

a. Prevent chafing
b. Secure wires and cables where routed through bulkheads and structural members
c. Fasten wires and cables in junction boxes, panels, and bundles for proper grouping and routing
d. Prevent mechanical strain or work hardening that would tend to break the conductors and connections
e. Prevent arcing or overheated wires and cables from causing damage to mechanical control cables

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f. Facilitate reassembly to equipment terminal boards
g. Prevent interference between wires and cables and other equipment
h. Prevent excessive movement in areas of severe vibration.

3.10.3.1 *Insulation.* Insulation tubing and tape shall be secured in place by tying or by clamps.

3.10.4 *Limitations.*

3.10.4.1 *Tapes.* Tapes that will dry out in service, produce chemical reaction with cable insulation, or absorb moisture shall not be used.

3.10.4.2 *Insulation tubing.* The use of insulation tubing to protect wires and cables from abrasion, chafing, exposure to fluids, and other conditions that may affect cable insulation shall be minimized. Nonrigid insulation tubing shall not be used for support of wires and cables.

3.10.4.3 *Moisture-absorbent-type materials.* Moisture-absorbent-type materials shall not be used as a fill for clamps or adapters.

3.10.4.4 *Wires and cables.* Wires and cables shall not be tied nor fastened together in conduit or insulation tubing.

3.10.4.5 *Cable supports.* Cable supports shall not restrict the wires or cables in such manner as to interfere with operation of shock mounts.

3.10.4.6 *Tape and cord.* Tape and cord shall not be used for primary support.

3.11 Conduit.

3.11.1 *Metallic conduit.* Upon approval of the procuring activity, metallic conduit may be used in the missile only where necessary to protect wiring or to facilitate maintenance in inaccessible portions of the missile. Metallic conduit shall be used, when necessary, between equipment and its related radio-noise filter unit. Metallic conduit may be used for shielding to meet the requirements of MIL–E–6051, subject to approval of the procuring activity.

3.11.1.1 *Rigid conduit.* Rigid aluminum or aluminum alloy tubing shall be in accordance with the sizes specified on AND10106. Rigid conduit utilizing other materials suitable for the purpose shall be used where advantages are to be gained by its use.

3.11.1.2 *Flexible conduit.* Flexible metallic conduit conforming to MIL–C–6136 shall be used only when the use of rigid metalic conduit is impracticable.

3.11.1.3 *Conduit fittings.* The use of fittings for metallic conduit shall be in accordance with AND10380, except as approved by the procuring activity for certain applications.

3.11.2 *Nonmetallic conduit.* Government-approved nonmetallic conduit, such as insulation tubing, may be used for electrical and mechanical protection of wiring, but shall not be installed in greater amounts than necessary.

3.11.2.1 *Types.* Nonmetallic conduit shall be of a material satisfactory for the ambient temperature encountered. Nonrigid electrical insulation material conforming to MIL–I–631, type F, grade b, form U, class 1, category 1; MIL–I–7444; or Government-approved equivalent shall be used for ambient temperatures within their specified capabilities or when installed in fuel tanks or pressurized compartments. Nonmetallic conduit approved by the procuring activity shall be used for higher temperatures.

3.11.2.2 *Fittings.* Government-approved devices shall be used for attaching and terminating nonmetallic conduit. Wherever practicable, AN fittings shall be used.

3.11.3 *Conduit installation.* Conduit shall be so installed as to withstand vibration and normal service abuse.

3.11.3.1 *Support.* Conduit shall be so installed as to relieve strain on the ferrules.

3.11.3.2 *Drainage.* Where practicable, conduit shall be so installed that fluids or condensate will not be entrapped. Suitable drainage holes shall be provided at the low points, except for metallic flexible conduit or when installed in fuel tanks or pressurized compartments. Burrs shall be removed from drainage holes and conduit terminations. Where drain holes cannot be provided, provisions shall be made to prevent entrapment of fluid in the conduit.

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3.11.3.3 *Grounding.* Grounding of metallic conduit shall be in accordance with MIL–B–5087.

3.11.3.4 *Raduis of bends.* Conduit and conduit fitting bends shall not cause chafing of cables.

3.11.3.5 *Diameter.* The group of wires and cables to be installed in the conduit shall be bundled together and the maximum diameter measured. The maximum diameter shall not exceed 80 percent of the internal diameter of the conduit. During the initial design, the maximum diameter cable in accordance with applicable specifications shall be used when making this measurement.

3.12 Maintainability.

3.12.1 *Accessibility.* Insofar as practicable, parts of the equipment requiring routine service checking, adjusting, or replacing shall be made readily accessible for servicing without teardown of the equipment and removal of any major part, component, or accessory.

3.12.2 *Assembly.* Insofar as practicable, the following general requirements shall be complied with in order to reduce the possibility of human errors during assembly and maintenance of system components.

3.12.2.1 Where lug-terminated electrical leads are required, one of the following methods shall be used (in order of preference):

a. Different lead lengths from the cable tie-down point
b. Different sized lugs
c. Identification of leads and terminals
d. Color coding of leads and terminals.

3.13 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The requirements of MIL–D–70327 shall govern changes in the manufacturer's part numbers.

3.14 Drawings.

3.14.1 *Wiring and interconnection diagrams.* Wiring and interconnection diagrams shall be prepared in accordance with MIL–D–70327 and submitted to the procuring activity as specified herein.

3.14.1.1 *Preliminary block diagrams and schematics (for Army and Air Force).* As specified in MIL–S–25063, preliminary block diagrams and schematics that show each electrical and electronic circuit shall be provided for each prototype missile. Presentation of circuits in straight line (elementary) form is acceptable and multiple sheets may be used.

3.14.1.2 *Schematic and interconnection diagrams (for Army and Air Force).* As specified in MIL–S–25063, schematic and interconnection diagrams, and any revisions thereto, for each type of missile shall be provided. These diagrams shall be on multiple sheets or in book form in accordance with MIL–H–5166. Each system, such as alternating-current power, direct-current power, exterior lighting, and engine control, shall be presented on one sheet, if practicable without undue crowding.

3.14.1.2.1 *Approval of schematic and interconnection diagrams (for Army and Air Force).* As specified in MIL–S–25063, schematic and interconnection diagrams shall be submitted to the procuring activity for approval prior to delivery of the first missile and subsequent to mockup approval (see 4.3).

3.15 Workmanship. Details of workmanship shall be in accordance with high-grade, missile, electrical wiring, and equipment installation practice and of such quality as to insure safety, proper operation, and service life. Details of workmanship shall be subject to the inspection and approval of the procuring activity.

3.15.1 *Loose parts.* All loose parts, chips, scraps, excess hardware, et cetera, shall be removed from the missile.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier's responsibility. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure

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supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of guided missile wiring installation shall be classified as acceptance tests.

4.3 Approval of schematic and interconnection diagrams. Prior to the scheduled date of delivery of the first missile, schematic and interconnection diagrams shall be submitted to the procuring activity for approval as specified in 3.14.1.2.1.

4.3.1 *Wiring mockup.* Unless otherwise specified (see 6.2), prior to delivery of the first missile, the contractor shall conduct a wiring mockup for inspection by the procuring activity. At this mockup, the contractor shall demonstrate typical wiring installations and specific wiring practices for which deviations are required.

4.3.2 *Inspection of first missile.* The wiring installation of the first complete missile shall be subject to approval by the procuring activity and shall be inspected at the contractor's plant by the contractor under the surveillance of the procuring activity.

4.3.3 *Changes in wiring.* A missile in which changes in the wiring have been made shall be inspected at the contractor's plant by the contractor under the surveillance of the procuring activity to determine compliance with the requirements of this specification.

4.3.4 *Individual tests.* Functional tests shall be conducted by the contractor on each missile to insure proper continuity of all electrical and electronic circuits and proper operation of all electrical and electronic equipment.

4.3.5 *Inspection.* Inspection shall be conducted on each missile by the contractor under surviellance of the procuring activity to insure compliance with the requirements of this specification.

5. PREPARATION FOR DELIVERY

5.1 This section is not applicable to this specification.

6. NOTES

6.1 Intended use. The installation of guided-missile wiring and wiring devices covered by this specification is intended to provide satisfactory interconnection of airborne electric and electronic missile equipment.

6.2 Ordering data. Procurement documents should specify the following:

a. If use of wires and cables of a type other than specified herein is desired (see 3.6.1.1, 3.6.1.2, and 3.6.1.3).
b. If ground cable letter N should not be used for identification (see table III, note 3, g(1)).
c. The number of preliminary block diagrams and schematics desired and the due date for submission to the procuring activity for approval (see 3.14.1.1). (Three copies are desirable. Sixty days after date of contract or letter of intent is considered a desirable due date.)
d. The number of schematic and interconnection diagrams desired and the due date for submission to the procuring activity for approval (see 3.14.1.2 and 3.14.1.2.1). (One reproducible and two prints are desirable. Sixty days after date of contractor or letter of intent is considered a desirable due data.)
e. If inspection of wiring mockup is not desired (see 4.3.1).

6.3 Definitions. For the purposes of this specification, the following definitions will apply:

a. Wires, electrical—A single, metallic conductor of solid, stranded, or tinsel construction designed to carry current in an electrical circuit; it may be bare or insulated, but does not have a metallic covering, sheath, or shield.
b. Cable, power, electrical—Two or more insulated conductors, solid or stranded, contained in a common covering; two or more insulated conductors twisted or molded together without common covering, or one insulated conductor with a metallic covering shield or outer conductor.
c. Wiring—An installation or the installing of wires and cables.

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d. Wiring devices—The accessory parts and materials that are used in the installation of wiring, such as terminals, connectors, junction boxes, conduit, clamps, insulation, and supports.

e. Flammable—The capability of bursting into flame when a spark or open flame is passed sufficiently near, as with fumes and vapors from hot oils or volatile combustible liquids and finely powdered combustible solids.

f. Guided missile—Any self-propelled, unmanned, space airborne vehicle designed primarily to strike a selected target.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:
Army—Ordnance
Navy—Bureau of Naval Weapons
Air Force

Preparing activity:
Air Force

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEM PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

FORM 1426
1 APR 63

REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

PLATE NO. 15419

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